

# MS Defender XDR Incidents API - Student Guide V4

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## Continuous Incident Generation Version

Welcome to the Educational Cybersecurity API V4! This guide will help you connect to and use the API for your course exam.

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## API Information

**Base URL:** `http://164.92.167.24`

**API Version:** 4.0 (Continuous Generation)

**Special Feature:** New security incidents are automatically generated every 2-60 minutes, simulating a real-world Security Operations Center (SOC) environment where threats are continuously detected.

**Note:** The `curl` and PowerShell examples throughout this guide are provided for quick testing and exploration only. Your exam project must be implemented in **Python** using the `requests` library.

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## Getting Started

### Step 1: Get Your Access Token

Before you can access the API, you need to obtain your personal access token.

#### curl Example:

```
curl -X POST http://164.92.167.24/api/auth/token \  
-H "Content-Type: application/json" \  
-d '{"email":"your.email@student.edu"}'
```

#### PowerShell Example:

```
$headers = @{"Content-Type"="application/json"}  
$body = @{"email"="your.email@student.edu"} | ConvertTo-Json  
$response = Invoke-RestMethod -Uri "http://164.92.167.24/api/auth/token" -Method  
POST -Headers $headers -Body $body  
  
Write-Host "Your token: $($response.token)"  
Write-Host "Expires in: $($response.expires_in_hours) hours"
```

#### Response:

```
{
  "message": "Token retrieved successfully",
  "email": "your.email@student.edu",
  "token": "student-xxxxxxxxxxxxx",
  "expires_at": "2026-05-12T20:00:00+00:00",
  "expires_in_hours": 48.0,
  "instructions": "Use this token in Authorization header: Bearer <token>",
  "note": "Token will expire in 48.0 hours. Request a new token if it expires."
}
```

**Important:**

- Your token expires after **48 hours**
- Save your token securely
- If your token expires, simply request a new one using the same email

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## Authentication

All API requests (except getting your token) require authentication using your token in the **Authorization** header.

**Format:**

```
Authorization: Bearer your-token-here
```

**curl Example:**

```
curl -H "Authorization: Bearer student-xxxxxxxxxxxxx" \
http://164.92.167.24/api/incidents
```

**PowerShell Example:**

```
$headers = @{"Authorization"="Bearer student-xxxxxxxxxxxxx"}
$response = Invoke-RestMethod -Uri "http://164.92.167.24/api/incidents" -Headers
$headers
$response
```

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## Available Endpoints

### 1. Get All Incidents

Retrieve a list of security incidents with optional pagination.

**Endpoint:** `GET /api/incidents`

**Query Parameters:**

| Parameter           | Description                                       | Example                |
|---------------------|---------------------------------------------------|------------------------|
| <code>\$top</code>  | Number of results to return (max 100, default 10) | <code>\$top=50</code>  |
| <code>\$skip</code> | Number of results to skip (for pagination)        | <code>\$skip=20</code> |

**curl Examples:**

```
# Get first 10 incidents (default)
curl -H "Authorization: Bearer YOUR_TOKEN" \
  http://164.92.167.24/api/incidents

# Get first 50 incidents
curl -H "Authorization: Bearer YOUR_TOKEN" \
  "http://164.92.167.24/api/incidents?$top=50"

# Pagination - skip first 20, get next 10
curl -H "Authorization: Bearer YOUR_TOKEN" \
  "http://164.92.167.24/api/incidents?$skip=20&$top=10"
```

**PowerShell Examples:**

```
$headers = @{"Authorization"="Bearer YOUR_TOKEN"}
$baseUrl = "http://164.92.167.24"

# Get first 10 incidents (default)
$response = Invoke-RestMethod -Uri "$baseUrl/api/incidents" -Headers $headers

# Get first 50 incidents
$response = Invoke-RestMethod -Uri "$baseUrl/api/incidents?$top=50" -Headers $headers
Write-Host "Total incidents: $($response.'@odata.count')"
```

```
Write-Host "Returned: $($response.value.Count) incidents"
```

```
# Display incidents
foreach($incident in $response.value) {
    Write-Host "$($incident.incidentId): $($incident.incidentName)"
}

# Pagination - skip first 20, get next 10
$response = Invoke-RestMethod -Uri "$baseUrl/api/incidents?$skip=20&$top=10" -
Headers $headers
```

**Response Structure:**

```
{
  "@odata.context": "https://api.security.microsoft.com/api/$metadata#Incidents",
  "@odata.count": 127,
  "@odata.nextLink": "/api/incidents?$skip=10&$top=10",
  "value": [
    {
      "incidentId": "INC1001",
      "incidentName": "APT33 - Phishing Campaign",
      "createdTime": "2026-04-10T20:30:00Z",
      "lastUpdateTime": "2026-04-10T22:15:00Z",
      "assignedTo": "soc.analyst@oresund-industrial.dk",
      "status": "Active",
      "severity": "High",
      "classification": "TruePositive",
      "determination": "Phishing",
      "tags": ["APT33", "Emotet", "Øresund Industrial"],
      "alerts": [...],
      "impactedEntities": {
        "users": 5,
        "machines": 2,
        "mailboxes": 1
      },
      "threatFamily": "Emotet",
      "mitreTechniques": ["T1566.001", "T1204.002"],
      "summary": "Security incident at Øresund Industrial involving APT33..."
    }
  ],
  "pagination": {
    "skip": 0,
    "top": 10,
    "returned": 10,
    "filtered_total": 127,
    "unfiltered_total": 127
  },
  "api_version": "4.0",
  "mode": "exam",
  "note": "New incidents are continuously generated in the background"
}
```

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## 2. Get Specific Incident

Retrieve detailed information about a single incident.

**Endpoint:** `GET /api/incidents/{incident_id}`

**curl Example:**

```
curl -H "Authorization: Bearer YOUR_TOKEN" \  
http://164.92.167.24/api/incidents/INC1001
```

### PowerShell Example:

```
$headers = @{"Authorization"="Bearer YOUR_TOKEN"}  
$incidentId = "INC1001"  
$response = Invoke-RestMethod -Uri  
"http://164.92.167.24/api/incidents/$incidentId" -Headers $headers  
  
Write-Host "Incident: $($response.incidentName)"  
Write-Host "Severity: $($response.severity)"  
Write-Host "Status: $($response.status)"  
Write-Host "Threat Actor: $($response.tags[0])"  
Write-Host "Number of alerts: $($response.alerts.Count)"  
  
# Examine alerts  
foreach($alert in $response.alerts) {  
    Write-Host "`nAlert: $($alert.title)"  
    Write-Host "    Severity: $($alert.severity)"  
    Write-Host "    Machine: $($alert.computerDnsName)"  
    Write-Host "    IPs: $($alert.entities.ips -join ', ')"  
    Write-Host "    Domains: $($alert.entities.domains -join ', ')"  
}
```

### Response Structure:

```
{  
  "incidentId": "INC1001",  
  "incidentName": "APT33 - Phishing Campaign",  
  "createdTime": "2026-04-10T20:30:00Z",  
  "lastUpdateTime": "2026-04-11T08:00:00Z",  
  "assignedTo": "soc.analyst@oresund-industrial.dk",  
  "severity": "High",  
  "status": "Active",  
  "classification": "TruePositive",  
  "determination": "Phishing",  
  "tags": ["APT33", "Emotet", "Øresund Industrial"],  
  "comments": [  
    {  
      "comment": "Initial analysis shows indicators consistent with APT33 TTPs  
targeting industrial manufacturing environments",  
      "createdBy": "soc.analyst@oresund-industrial.dk",  
      "createdTime": "2026-04-10T21:30:00Z"  
    }  
  ],  
  "alerts": [  
    {
```

```
"alertId": "ALT1001-0",
"title": "Suspicious PowerShell execution detected on SCADA01-F1",
"category": "Malware",
"severity": "High",
"detectionSource": "Microsoft Defender for Endpoint",
"machineId": "SCADA01-F1",
"computerDnsName": "SCADA01-F1.oresund-industrial.dk",
"entities": {
  "ips": ["192.168.20.10", "185.220.100.240"],
  "domains": ["update-oresund.com"],
  "fileHashes": ["d41d8cd98f00b204e9800998ecf8427e"],
  "emails": ["eng.larsen@oresund-industrial.dk"],
  "processes": ["powershell.exe", "rundll32.exe"]
},
"firstActivity": "2026-04-10T20:35:00Z",
"lastActivity": "2026-04-11T06:00:00Z"
},
],
"mitreTechniques": ["T1566.001", "T1204.002"],
"impactedEntities": {"users": 5, "machines": 2, "mailboxes": 1},
"threatFamily": "Emotet",
"summary": "Security incident at Øresund Industrial Manufacturing A/S involving APT33..."
}
```

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## Rate Limits

The API enforces rate limits:

- **50 requests per minute**
- **1,500 requests per hour**

If you exceed these limits, you'll receive a **401 Unauthorized** response.

### Best Practice - Add delays:

```
# Wait 5 minutes between requests
time.sleep(300)
```

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## Understanding the Data

### Incident Structure

Each incident contains:

- **incidentId**: Unique identifier (e.g., "INC1001")
- **incidentName**: Descriptive name combining threat actor and attack type
- **createdTime**: When the incident was first detected (ISO 8601 format)

- **lastUpdateTime**: When the incident was last updated
- **assignedTo**: Analyst assigned to the incident (can be null)
- **severity**: Low, Medium, High, or Critical
- **status**: Active, InProgress, or Resolved
- **classification**: TruePositive, FalsePositive, or InformationalExpectedActivity
- **determination**: Type of threat (e.g., Malware, Phishing, MultiStagedAttack)
- **tags**: Array containing threat actor and malware family names
- **comments**: Array of analyst comments with text, author, and timestamp
- **alerts**: Array of related security alerts (see Alert Structure below)
- **impactedEntities**: Count of affected users, machines, and mailboxes
- **threatFamily**: Name of the malware family involved
- **mitreTechniques**: Array of MITRE ATT&CK technique IDs
- **summary**: Text description of the incident

## Alert Structure

Each alert within an incident contains:

- **alertId**: Unique alert identifier
  - **title**: Descriptive alert title
  - **category**: Type of threat (Malware, Phishing, Lateral Movement, etc.)
  - **severity**: Alert severity level (Low, Medium, High, Critical)
  - **detectionSource**: Microsoft product that detected the alert
  - **machineId**: Identifier of the affected machine
  - **computerDnsName**: DNS name of the affected machine
  - **entities**: Object containing Indicators of Compromise (IOCs):
    - **ips**: Array of IP addresses
    - **domains**: Array of domain names
    - **fileHashes**: Array of file hash values
    - **emails**: Array of email addresses
    - **processes**: Array of process names or command lines
    - **urls**: Array of URLs (present on certain incidents)
  - **firstActivity**: Timestamp of first observed activity
  - **lastActivity**: Timestamp of last observed activity
- 

## What's New in V4?

### Continuous Incident Generation

Unlike earlier versions with a fixed dataset, V4 continuously generates new incidents:

- **Startup**: Begins with 25 initial incidents
- **Background Generation**: 1-3 new incidents created every 2-60 minutes
- **Growth Rate**: Approximately 90+ new incidents per day
- **Course Duration**: Expected to reach 2,600-2,800 incidents over 4 weeks

### Why is this important?

- Simulates a real Security Operations Center where threats appear continuously
- Allows you to practice monitoring for new threats over time
- Enables time-based analysis and trending
- More realistic incident response scenarios

## Øresund Industrial Context

All incidents are scoped to **Øresund Industrial Manufacturing A/S**. Machine names, user accounts, IPs, and domains all reflect this organisation. Expect hostnames like `SCADA01-F1.oresund-industrial.dk`, `ERP01-TCORP.oresund-industrial.dk`, and users such as `eng.larsen@oresund-industrial.dk`.

## SQL Injection Awareness

Some incidents contain realistic attacker-captured payloads in `processes` and `urls` entity fields (e.g., sqlmap command lines and malicious HTTP requests). When storing these values in your own database, always use **parameterised queries** — never string concatenation — to avoid executing attacker SQL against your own database.

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## Extras

These features are not required for the exam project but are available if you want to explore further.

### Filtering Incidents

The `/api/incidents` endpoint accepts optional query parameters to filter results server-side. This can be useful for quick exploration but is **not a substitute for querying your own database** in the project.

| Parameter                 | Description                                 | Example                         |
|---------------------------|---------------------------------------------|---------------------------------|
| <code>severity</code>     | Filter by severity level                    | <code>severity=High</code>      |
| <code>status</code>       | Filter by incident status                   | <code>status=Active</code>      |
| <code>threat_actor</code> | Filter by threat actor name (partial match) | <code>threat_actor=APT33</code> |

### curl Example:

```
curl -H "Authorization: Bearer YOUR_TOKEN" \  
  "http://164.92.167.24/api/incidents?severity=Critical&\$top=50"
```

### PowerShell Example:

```
$headers = @{"Authorization"="Bearer YOUR_TOKEN"}  
$response = Invoke-RestMethod -Uri "http://164.92.167.24/api/incidents?  
severity=Critical&\$top=50" -Headers $headers  
Write-Host "Critical incidents: $($response.'@odata.count')"
```



### 3. Get Incident Summary

Returns aggregated statistics across all incidents — useful for a quick overview without fetching the full dataset.

**Endpoint:** `GET /api/incidents/summary`

**curl Example:**

```
curl -H "Authorization: Bearer YOUR_TOKEN" \  
http://164.92.167.24/api/incidents/summary
```

**PowerShell Example:**

```
$headers = @{"Authorization"="Bearer YOUR_TOKEN"}  
$response = Invoke-RestMethod -Uri "http://164.92.167.24/api/incidents/summary" -  
Headers $headers  
  
Write-Host "Total incidents: $($response.total_incidents)"  
Write-Host "Recent (last 7 days): $($response.recent_incidents)"  
$response.by_severity  
$response.top_threat_actors
```

**Response Structure:**

```
{  
  "total_incidents": 312,  
  "recent_incidents": 47,  
  "by_severity": {  
    "Critical": 28,  
    "High": 105,  
    "Medium": 134,  
    "Low": 45  
  },  
  "by_status": {  
    "Active": 120,  
    "InProgress": 89,  
    "Resolved": 103  
  },  
  "by_classification": {  
    "TruePositive": 198,  
    "FalsePositive": 67,  
    "InformationalExpectedActivity": 47  
  },  
  "top_threat_actors": {  
    "APT33": 42,  
    "Lazarus Group": 38,  
    "Volt Typhoon": 31  
  }  
}
```

```
    },  
    "top_malware_families": {  
      "Cobalt Strike": 55,  
      "Emotet": 48,  
      "TrickBot": 41  
    }  
  }  
}
```

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## Common Issues

- Issue: "Invalid or expired token"
  - **Solution:** Request a new token using your email address:
- Issue: "Rate limit exceeded"
  - **Solution:** Add delays between requests
- Issue: No new incidents appearing
  - **Solution:** V4 generates incidents every 2-60 minutes randomly. The background generation is automatic. Check back later or use the summary endpoint to see current total

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Good luck with your exam!